

Supplementary information for CSPG4 tissue RNA in extraskeletal myxoid chondrosarcoma: an exploratory comparison across sarcoma cohorts

The scientific measurements, frozen protocols and original reproducibility archive are unchanged from Research Square version 1 (<https://doi.org/10.21203/rs.3.rs-10959636/v1>). The complete public code/data archive is <https://assets-eu.researchsquare.com/files/rs-10959636/v1/b95606f3a3799c60c8ac4434.zip>. Its SHA256 is 56e089a6b39b9d5d5140a4185987778803359ab1154d26beb882d6e6197f3ea6. File names below identify its preserved components; no new analysis or public preregistration is claimed.

Source inventory and specimen scope

This supplement records which specimens were included, what each assay measured, and how the comparisons can be reproduced. EMC means extraskeletal myxoid chondrosarcoma. The comparator abbreviations are MLPS (myxoid liposarcoma), LGFMS (low-grade fibromyxoid sarcoma), MFS (myxofibrosarcoma), DFSP (dermatofibrosarcoma protuberans) and SFT (solitary fibrous tumor). HPA is the Human Protein Atlas. TPM means transcripts per million; RMA is the robust multi-array average method used for the released array signals. IHC means immunohistochemistry, and ICC/IF means immunocytochemistry/immunofluorescence. These antibody-based assays provide context on protein staining and location; they are distinct from the RNA measurements.

Source	Units and usable specimens	Assay / release	Role
Hofvander 2026	704 source patients; 9 primary EMC after fixed exclusions	19,116 gene symbols, gene TPM; author release v1.0.1	New-to-this-analysis tissue cohort
GSE24369	42 original arrays: 6 EMC, 17 LGFMS, 6 MFS, 6 desmoid, 5 SFT, 2 muscle pools	GPL6244 transcript-cluster RMA log2 signal	Retrospective same-histology anchor; also CHRNA6 discovery data
GSE28866	Four EMC library/STT records; individual patient count unverified	Selected 3SEQ peaks, supplied compressed normalized scale	Historical, different-estimand sample/organ context only
HPA	12 gene XML/JSON pairs; tissue/cell records are not independent subjects	XML version 25; current methods/download 25.1, Ensembl 109	Descriptive normal RNA, IHC and ICC/IF context
Cammareri 2023	Five EMC rows in original supplementary workbook	Whole-section PRAME QR005 IHC; both reader columns negative	External negative protein evidence, not paired with present RNA

Nine Hofvander EMC specimens met the fixed inclusion policy: 11881-19, 3371-22, 3372-22, 4716-22, 4840-13, 5149-18, 5241-06, 7931-19 and 8102-22. The excluded previously reported cases 104-92,168-97 and 536-00 map to source MDB9736 cases 3, 4 and 7. The fourth excluded specimen, 5081-14, is a local recurrence. For all 704 source samples, the records preserve the original Table S1 diagnosis and any nonblank specimen exception.

Specimen identifiers do not resolve all questions about provenance. No patient date is inferred from an accession suffix, and blank publication references do not prove independence. A gene-fusion-positive result in this source may come from current RNA sequencing or an earlier assay. It does not necessarily identify the fusion partner in every case.

Original-diagnosis primary comparators contain 14 MLPS, 13 LGFMS, 18 synovial sarcomas; separate contexts contain 60 MFS and 10 DFSP. Shared-histology secondary contexts add 11 primary SFT and 11 desmoids. Source metadata sometimes uses revised or transcriptomic labels; original labels are primary and

the separately reported revision sensitivity changes all relevant rows symmetrically. Purity and grade are unavailable for EMC adjustment.

Only four of the nine EMC patients contribute to the primary comparisons within sequencing years. The nine EMC specimens were sequenced in 2019(n=2), 2020(n=1), 2021(n=1), 2022(n=2) and 2023(n=3). MLPS has 2 comparators in 2019 and 5 in 2021; LGFMS has 7 and 5. Synovial sarcoma has 10 in 2019 and 8 in 2020. Each primary matched comparison therefore contains 3 EMC, with 4 unique EMC patients across the comparisons. The separate DFSP comparison has only 1 matched EMC and 3 comparators in 2021. Year groups without both relevant specimen types remain null. All counts refer to the fixed specimen policy, not the full published histology totals.

Removing the 2019 sample group leaves seven EMC patients in the marginal analysis, but only two contribute to the primary year-matched comparisons. The 2021 patient contributes to the MLPS and LGFMS comparisons, and the 2020 patient contributes to the synovial sarcoma comparison. Thus year removal changes the patients and comparator groups supporting the estimate; it does not establish a biological or technical effect of sequencing year.

Exact measurement contract

TPM source SHA256: b0d665d1bd1d96ace1faf66cc5a4d7ab7e41cb487c8f0f61734f102a1f9a7af3.

Original GSE24369 family gzip SHA256:

98c83c8ca23b7052cf0d4d0099a7bf1af6c3c972276038c3a633e2a5349b3c37. Full source digests and byte sizes are in the two frozen metadata manifests. Analysis uses released numeric precision, not the earlier rounded cache.

Gene	GPL6244 probe	Role
CD276	7984743	Address
SSTR2	8009526	Address
PRAME	8074856	Address
FAP	8056257	Address
CD248	7949588	Address
CSPG4	7990545	Address
MSLN	7992071	Address
L1CAM	8175871	Address
GPC3	8175234	Address
ALPP	8049123	Address
CDH17	8151795	Address
CHRNA6	8150550	Separate context control

Each selected array probe maps consistently to one gene across all of its transcript-cluster assignments. We do not average probes, choose a best probe, substitute a gene alias or filter features according to expression. Finite signed log values are valid array signals; TPM must be finite and nonnegative. Zero observations remain ties. Missing or duplicate observations trigger a recorded technical failure instead of deletion or imputation. The actual runs completed without these failures.

The score A averages pairwise comparisons of RNA values. For EMC values x_i and comparator values y_j , $A = (1/nm) \sum_i \sum_j [I(x_i > y_j) + 0.5I(x_i = y_j)]$. Here, n and m are the numbers of EMC and comparator specimens, and I equals 1 when its stated condition is true and 0 otherwise. Tied pairs contribute one half. Pair counts are not independent-patient counts.

The marginal summary is the mean of the 3 histology scores. For the matched summary, we first calculate the score within each exact year. Within each histology, we then weight these year scores by the number of supported EMC specimens, before averaging equally across histologies. A common EMC specimen can contribute to several histology comparisons. Bootstrap sampling preserves that dependence by sharing the specimen's resample across comparisons. No cross-platform scale or patient pooling is performed.

Bootstrap resampling used seed 20260906 and 2000 replicates within histology/year groups. The conditional 95% percentile intervals are pointwise: each describes one estimate rather than simultaneous coverage of the full family of estimates. They were not used to select genes. For CSPG4, the MLPS matched interval has identical endpoints, [0.66667, 0.66667]. This reflects sparse observed groups and the resampling design, not absence of biological uncertainty. The files identify all single-specimen groups and every patient's placement. Independent 5000-draw resampling reproduced the main matched interval endpoints. That numerical check does not remove small-sample limitations.

The original operational rule selects genes meeting all three conditions: marginal composite ≥ 0.70 ; each primary marginal/matched histology >0.5 ; and both summaries >0.5 after each single EMC and histology deletion. Removing sequencing years or comparator specimens and using revised diagnoses qualify interpretation; they are not extra pass criteria. After a deletion, the remaining supported-year weights are normalized again. A lost comparison remains undefined. Complete-case selection or imputation is not performed. No new replication success cutoff was added. The separate cross-cohort output compares score direction and size against LGFMS primarily, and MFS/SFT/desmoid secondarily.

The rule is retained as specified, including every pass and fail result. It is not a validated therapeutic ranking and does not exclude other genes from biological follow-up. The revised manuscript emphasizes descriptive comparisons after review of the completed analysis; neither the original protocol nor the computations have been changed. A compares groups within a gene and does not measure absolute expression differences between genes.

Tables and evidence files

Table S1, `../../../../autonomy/atlas-hofvander-validation-2026-09-06/all12-gene-effects.csv`, includes every gene, both primary composites, conditional intervals and same-LGFMS effects. Table S2, `../../../../autonomy/atlas-hofvander-validation-2026-09-06/all-hofvander-contrasts.csv`, contains 60 gene $\times$ histology rows; `../../../../autonomy/atlas-hofvander-validation-2026-09-06/all-year-cells.csv` provides all supported/unsupported cells. Table S3, `../../../../autonomy/atlas-hofvander-validation-2026-09-06/all-shared-histology-replication.csv`, contains 48 gene $\times$ shared-histology comparisons and individual-deletion ranges. `../../../../autonomy/atlas-hofvander-validation-2026-09-06/all-primary-deletions.csv` retains every summary deletion. Full JSON adds every specimen value, placement, individual contrast and revised-label mapping.

Table S4, `../../../../autonomy/atlas-hofvander-validation-2026-09-06/draft/normal-context.csv`, contains 12 source-linked HPA records with normal IHC reliability/missingness and assay-location context. It is a descriptive extract, not a normal-sparing assessment. Source data include 1092 RNA tissue/block records and 1054 categorical normal-IHC cell records; neither total is a count of newly measured independent donors. HPA consensus RNA summarizes maxima across HPA/GTEx sources/subtissues. GPC3/CHRNA6 lack tissue-IHC rows in this recovery; absence of a row is not a negative stain. SSTR2/FAP uncertain IHC and ALPP multi-gene antibody warning remain explicit. PRAME's localization tags are not evidence of endogenous EMC peptide-HLA presentation.

PRAME supplementary workbook SHA256:

20b4bd1243be22e4eee2da7470da986091f91cc4959fa416708748ba99187ca7

Sheet 1 rows 128–132 record five EMC specimens, all negative in both reader columns. Blank percentage fields were not converted to measurements. Case 2 has EWSR1::NR4A3 RNA-sequencing support. Cases 3–

5 are NR4A3-positive by fluorescence in situ hybridization (FISH), while case 1 has no recorded ancillary method. These observations support a negative result in those 5 specimens, not universal PRAME absence or absent peptide-HLA presentation.

Reproduction and figure status

Empirical programs are `../../autonomy/atlas-hofvander-validation-2026-09-06/analyze.py` and `../../autonomy/atlas-hofvander-validation-2026-09-06/replication.py`; all 8 pre-outcome approved files are listed in `../../autonomy/atlas-hofvander-validation-2026-09-06/coordinator-authorization.json`. Original scientific protocol bytes are preserved alongside frontmatter-corrected versions. Each analysis ran once: command elapsed 46.38 and 12.72 seconds, exit 0. Runtime was Python 3.12.14, NumPy 2.3.5, openpyxl 3.1.5; exact executable/version receipt is in `../../autonomy/atlas-hofvander-validation-2026-09-06/draft/runtime.json`. Independent original-source arithmetic reproduced 8,448 TPM values, 504 array values, 2,436 estimate/deletion blocks and 266,772 scalar comparisons without discrepancy. These checks verify computation, not causal or clinical validity.

Source locations in the original manifests remain historical workstation paths. The portable offline wrapper accepts an explicit bundle root, stages exact archive/gzip members and changes only `source_location` in derived manifests. One replay in a fresh directory on the same Windows host completed with 829,482 exact scalar comparisons and zero discrepancies across 29 scientific JSON files and five CSV exports. Two execution records separately matched status and final stage. This is not a second-machine or second-OS claim. Python 3.12.14, NumPy 2.3.5, openpyxl 3.1.5 and et-xmlfile 2.0.0 are pinned. Historical approval is preserved, while the compatibility record explicitly identifies the mechanical replay. Installation instructions suffice; no bundled runtime or wheels are claimed.

The canonical `plot_emc_tissue_rna.py` adapter preserves the frozen draft plotting statistics and geometry. It adds SVG output for the manuscript renderer alongside vector PDF and 300-dpi PNG. Rendering is rejected until the exact source and plotting-script bytes are committed. Figures remain descriptive; no uncertainty bars or new statistical tests are introduced. Normal-expression context remains Table S4. Figure rendering and visual inspection are recorded separately from numerical replay.

Author identity, affiliation, correspondence and ORCID match the posted Research Square article. The author reports no funding and no competing financial interests. One non-financial interest is declared: the author is a survivor of extraskeletal myxoid chondrosarcoma, the disease this work addresses. This public-data reanalysis asserts no new ethics approval, exemption or waiver. OpenAI Codex assisted with implementation, source organization, verification and drafting; this assistance is disclosed in the main article.

The historical analysis, draft and replay packets are preserved unchanged. Canonical manuscript routing and the SVG plotting adapter are separate presentation files; no scientific decision or source membership changed.

Table S1 All fixed panel effects

A=0.5 means neither group tends to have higher RNA levels. The CHRNA6 control is separate from the candidate-gene pass count. Matched columns use only the supported-year specimens, not all nine EMC. The machine-readable tables described above retain the intervals and every sensitivity calculation.

Gene	Hofvander marginal A	Hofvander matched A	LGFMS array A	LGFMS Hofvander marginal A	Rule
CD276	0.26575	0.27143	0.32353	0.02564	Fail
SSTR2	0.56926	0.73373	0.47059	0.76068	Fail
PRAME	0.21573	0.25397	0.68627	0.64103	Fail
FAP	0.34367	0.27897	0.54902	0.05128	Fail

Gene	Hofvander marginal A	Hofvander matched A	LGfMS array A	LGfMS Hofvander marginal A	Rule
CD248	0.12178	0.01587	0.30392	0.21368	Fail
CSPG4	0.89454	0.81111	1.00000	0.96581	Pass
MSLN	0.68008	0.58968	0.29412	0.92735	Fail
L1CAM	0.69931	0.61627	0.76471	0.90171	Fail
GPC3	0.29060	0.31270	0.48039	0.76068	Fail
ALPP	0.53336	0.49722	0.63725	0.61111	Fail
CDH17	0.44520	0.43115	0.09804	0.30342	Fail
CHRNA6	1.00000	1.00000	1.00000	1.00000	Control only